

EE446 Lab 5: Questions to Answer

Aidan Goodman

July 2026

1 Questions

1. Purpose of Spectrogram/MFCC Features over Raw Waveforms

When we have raw audio data, it is traveling through time which makes it difficult feed into a keyword spotting model. This is because the keyword can occur at any-point through the audio recording, meaning that the same keyword can look totally different depending on the start and end times. We can convert the raw time-dependent data into spectrograms and MFCCs to better visualize features of the audio data to feed into a CNN. Additionally, the MFCC compression allows us to use a lower computation threshold to process the same data.

2. Purpose of the Representative Dataset in INT8 Quantization

We must use a representative dataset in INT8 quantization to insure that the model still understands the dynamic range of the activation thresholds. We must make sure that the model has accurately mapped these values to the 8 bit integer space without a major loss of information and accuracy.

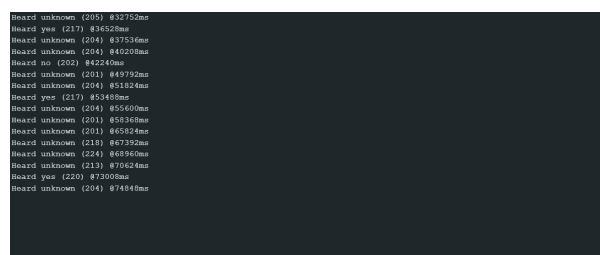
3. Function of `audio_processor.get_data()`

This function handles the loading, partitioning and preparation of the audio files for the model. It places them into the correct division with ground truth labels. You are allowed to set effects such as how much of the audio is background noise, and how many words we would like to classify.

4. Role of `g_model` and `g_model_len`

These need to be copied currently, since they represent the data of the quantized TensorFlow Lite model, both architecture and weights. The ending array size is the total amount of data for the model. The microcontroller can read this data in the memory for fast deployment and processing. .

5. Keyword Identification Screenshot



```
Heard unknown (205) 032752ms
Heard yes (217) 036520ms
Heard unknown (204) 037516ms
Heard unknown (204) 040200ms
Heard no (202) 042240ms
Heard unknown (201) 049792ms
Heard unknown (204) 051224ms
Heard yes (217) 053468ms
Heard unknown (204) 055600ms
Heard unknown (201) 058368ms
Heard unknown (201) 060368ms
Heard unknown (218) 067392ms
Heard unknown (224) 068960ms
Heard unknown (213) 070624ms
Heard yes (220) 073008ms
Heard unknown (204) 074848ms
```

Figure 1: Deployment Serial Monitor